MATHS MANTHAN

• TUITION CLASSES •

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Subject: Mathematics | **CBSE CLASS 10 – PERIODIC TEST 10 (SOLUTIONS)** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 5: Arithmetic Progressions** | **MM: 30**

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, all intermediate algebraic steps, and clearly boxed final answers.
2. In internal choice (**OR**) questions, both options are solved completely with identical step-marking rigor.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 1.5M, 2M, 3M]**.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason MATHS MANTHAN | (5 × 1 = 5 Marks)

Q1. Which term of the AP $3, 8, 13, 18, \ldots$ is 78? [1]

(A) 14th (B) 15th (C) 16th (D) 17th

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(C) 16th** [0.5 Mark]

Given: $a = 3$, $d = 8 - 3 = 5$, $a_n = 78$.

Formula / Theorem Used: $a_n = a + (n-1)d$

Justification: [0.5 Mark]

$3 + (n-1)(5) = 78 \implies 5(n-1) = 75 \implies n - 1 = 15 \implies n = \mathbf{16}$.

Q2. How many terms are there in the finite AP $7, 13, 19, \ldots, 205$? [1]

(A) 33 (B) 34 (C) 35 (D) 36

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(B) 34** [0.5 Mark]

Given: $a = 7$, $d = 13 - 7 = 6$, $a_n = 205$.

Formula / Theorem Used: $a_n = a + (n-1)d$

Justification: [0.5 Mark]

$7 + (n-1)(6) = 205 \implies 6(n-1) = 198 \implies n - 1 = 33 \implies n = \mathbf{34}$.

Q3. In an AP, if $a = 2$ and $d = 4$, then the sum of its first 40 terms (S_{40}) is: [1]

(A) 3200 (B) 3280 (C) 3240 (D) 3300

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(A) 3200** [0.5 Mark]

Given: $a = 2$, $d = 4$, $n = 40$.

Formula / Theorem Used: $S_n = \frac{n}{2}[2a + (n-1)d]$

Justification: [0.5 Mark]

$S_{40} = \frac{40}{2}[2(2) + (40-1)(4)] = 20(4 + 39 \times 4) = 20(4 + 156) = 20 \times 160 = \mathbf{3200}$.

Q4. If the nth term of an AP is $a_n = 3n + 7$, then its common difference d is: [1]

(A) 3 (B) 7 (C) 10 (D) 4

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 1 Mark]

Correct Answer: (A) 3 [0.5 Mark]

We evaluate $d = a_2 - a_1$:

Formula / Theorem Used: $d = a_2 - a_1$

Justification: [0.5 Mark]

$a_1 = 3(1) + 7 = 10$ and $a_2 = 3(2) + 7 = 13 \implies d = 13 - 10 = \mathbf{3}$.

Q5. **Assertion (A):** The sum of the first n odd natural numbers is n^2.
Reason (R): The first n odd natural numbers form an AP with $a = 1$, $d = 2$, yielding $S_n = n^2$. [1]

(A) Both (A) and (R) are true, and (R) is correct explanation.

(B) Both (A) and (R) are true, and (R) is not correct explanation.

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 1 Mark]

Correct Answer: (A) Both (A) and (R) are true, and (R) explains (A). [0.5 Mark]

Evaluating $S_n = \frac{n}{2}[2a + (n-1)d]$ for $a = 1, d = 2$:

Formula / Theorem Used: $S_n = \dfrac{n}{2}\left[2(1) + (n-1)(2)\right] = \dfrac{n}{2}(2n) = n^2$

Justification: [0.5 Mark]

Both (A) and (R) are mathematically true, and (R) gives the exact AP derivation explaining (A).

SECTION B: Very Short Answer Questions — MATHS MANTHAN | (2 × 2 = 4 Marks)

Q6. If the 14th term of an AP is twice its 8th term and its 6th term is -8, find a and d. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given: $a_{14} = 2a_8$ and $a_6 = -8$.

Formula / Theorem Used: $a_n = a + (n-1)d$

Step 1: Translating $a_{14} = 2a_8$ [1 Mark]

$$a + 13d = 2(a + 7d) \implies a + 13d = 2a + 14d \implies a = -d \quad \text{— (i)}$$

Step 2: Substituting into $a_6 = -8$ [1 Mark]

$a + 5d = -8$. Replacing $a = -d$: $-d + 5d = -8 \implies 4d = -8 \implies d = -2$.
Thus $a = -(-2) = 2$.

$$\boxed{\mathbf{a = 2}, \qquad \mathbf{d = -2}}$$

Q7. Find the value of p so that $2p - 1$, 7, and $3p$ are three consecutive terms of an AP. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Given: $2p - 1$, 7, and $3p$ are consecutive terms of an AP.

Formula / Theorem Used: If a, b, c are in AP, then $2b = a + c$

Step 1: Forming Equation [1 Mark]

$2(7) = (2p - 1) + 3p$.

Step 2: Solving for p [1 Mark]

$14 = 5p - 1 \implies 5p = 15 \implies \boxed{\mathbf{p = 3}}$ (Terms: $5, 7, 9$).

SECTION C: Short Answer Questions — MATHS MANTHAN | (2 × 3 = 6 Marks)

Q8. How many three-digit numbers are divisible by 9? Also find their sum. [3]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Identifying the AP and Number of Terms n [1.5 Marks]

Three-digit numbers range from 100 to 999. First multiple of 9 is 108 (9×12) and last is 999 (9×111).
AP: $108, 117, 126, \ldots, 999$ ($a = 108$, $d = 9$, $a_n = 999$).

Formula / Theorem Used: $a_n = a + (n-1)d$ and $S_n = \frac{n}{2}(a + a_n)$

$$108 + (n-1)(9) = 999 \implies 9(n-1) = 891 \implies n - 1 = 99 \implies n = 100$$

Step 2: Computing the Sum S_{100} [1.5 Marks]

$$S_{100} = \frac{100}{2}(108 + 999) = 50 \times 1{,}107 = 55{,}350$$

$$\boxed{\mathbf{n = 100}, \qquad \mathbf{S_{100} = 55{,}350}}$$

Q9. The four angles of a convex quadrilateral are in AP. If the greatest angle is twice the least angle, find all four angles. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Given: Let angles be a, $a+d$, $a+2d$, $a+3d$ ($d>0$). Angle sum of a quadrilateral $= 360°$.

Formula / Theorem Used: Angle sum: $a+(a+d)+(a+2d)+(a+3d)=360°$

Step 1: Setting Up Both Equations [1.5 Marks]

$4a+6d=360° \implies 2a+3d=180°$ — (i). Given greatest $= 2\times$ least: $a+3d=2a \implies a=3d$ — (ii).

Step 2: Solving for d, a, and Listing Angles [1.5 Marks]

Substituting $a=3d$ into (i): $2(3d)+3d=180° \implies 9d=180° \implies d=20°$, so $a=3(20°)=60°$.

Angles: $60°$, $60°+20°=80°$, $60°+40°=100°$, $60°+60°=120°$.

Four angles: **60°**, **80°**, **100°**, **120°**

SECTION D: Long Answer Questions — MATHS MANTHAN | (2 × 5 = 10 Marks)

Q10. The houses of a row are numbered consecutively from 1 to 49. Show that there is a value of x such that the sum of the numbers of the houses preceding the house numbered x is equal to the sum of the numbers of the houses following it. Find this value of x. **[5]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Step 1: Formulating Preceding and Following House Sums [1.5 Marks]

House numbers $1,2,3,\ldots,49$ form an AP with $a=1$, $d=1$.

Houses preceding house x: $1,2,\ldots,(x-1)$, with sum S_{x-1}.

Houses following house x: $(x+1),(x+2),\ldots,49$, with sum $S_{49}-S_x$.

Formula / Theorem Used: Condition: $S_{x-1}=S_{49}-S_x$, where $S_k=\dfrac{k(k+1)}{2}$

Step 2: Setting Up the Equation in x [2 Marks]

$$\frac{(x-1)[(x-1)+1]}{2}=\frac{49(49+1)}{2}-\frac{x(x+1)}{2}$$

Multiplying both sides by 2:

$$(x-1)x=49\times 50-x(x+1) \implies x^2-x=2{,}450-x^2-x$$

Step 3: Solving for x and Validating Bounds [1.5 Marks]

$$x^2+x^2-x+x=2{,}450 \implies 2x^2=2{,}450 \implies x^2=\frac{2{,}450}{2}=1{,}225 \implies x=\sqrt{1{,}225}=35$$

(Rejecting $x=-35$ since house numbers satisfy $1\le x\le 49$.) Verification: $S_{34}=\frac{34\times 35}{2}=595$ and $S_{49}-S_{35}=1{,}225-630=595$.

x = 35

Q11. If the pth, q^{th}, and r^{th} terms of an AP are a, b, c respectively, prove that:

$$a(q-r)+b(r-p)+c(p-q)=0$$

[5]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Step 1: Representing a, b, c Using Distinct AP Parameters [1.5 Marks]

Let the first term be A and common difference be D (to avoid symbol clash with a).

Formula / Theorem Used: $T_n = A + (n-1)D$

$a = T_p = A + (p-1)D, \quad b = T_q = A + (q-1)D, \quad c = T_r = A + (r-1)D.$

Step 2: Substituting Into LHS and Grouping Coefficients of A and D [2 Marks]

$$\begin{aligned}\text{LHS} &= [A+(p-1)D](q-r) + [A+(q-1)D](r-p) + [A+(r-1)D](p-q) \\ &= A[(q-r)+(r-p)+(p-q)] + D[(p-1)(q-r)+(q-1)(r-p)+(r-1)(p-q)]\end{aligned}$$

Step 3: Expanding and Cancelling Pairwise Terms [1.5 Marks]

Coefficient of A: $(q-q)+(r-r)+(p-p)=0$.

Coefficient of D:

$$(pq-pr-q+r)+(qr-pq-r+p)+(rp-qr-p+q)=0$$

Thus LHS $= A(0) + D(0) = \boxed{\mathbf{0} = \text{RHS} \quad \text{[Hence Proved]}}$.

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Q12. CASE STUDY: Eco-Club School Tree Plantation Campaign

During a school environmental drive, students from Class I to Class XII plant saplings around the campus to curb air pollution. It is decided that each section of each class will plant as many trees as the class number in which they study (e.g., 1 tree per section of Class I, 2 trees per section of Class II, up to 12 trees per section of Class XII). Every class from I to XII has 3 sections. [5]

(i) How many trees in total are planted by Class I? [1]

(ii) How many trees in total are planted by Class XII? [1]

(iii) Calculate the grand total number of trees planted by students of Classes I to XII. [3]

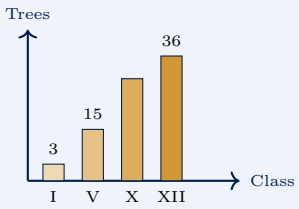
— **OR** —

Find the total number of trees planted exclusively by the primary wing (Classes I to V). [3]

▶ **CASE-BASED SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Extracted Scenario Data: Class k $(1 \leq k \leq 12)$ with 3 sections plants $3k$ trees, forming the AP $3, 6, 9, \ldots, 36$ with $a = 3, d = 3, n = 12$.

Formula / Theorem Used: $a_k = 3k$ and $S_n = \frac{n}{2}(a + a_n)$



Sub-part (i): Trees Planted by Class I [1 Mark]

With 3 sections planting 1 tree each: $a_1 = 3 \times 1 = \boxed{\mathbf{3}\text{ trees}}$.

Sub-part (ii): Trees Planted by Class XII [1 Mark]

With 3 sections planting 12 trees each: $a_{12} = 3 \times 12 = \boxed{\mathbf{36}\text{ trees}}$.

Sub-part (iii): Total Trees Planted by Classes I to XII [3 Marks]

Summing the 12-term AP $3, 6, 9, \ldots, 36$ $(n = 12, a = 3, l = 36)$:

$$S_{12} = \frac{12}{2}(3 + 36) = 6 \times 39 = \boxed{\mathbf{234}\text{ trees}}$$

Sub-part (iii) [OR Option]: Total Trees Planted by Primary Wing (Classes I to V) [3 Marks]

Here $n = 5$, $a = 3$ (Class I), and $a_5 = 3 \times 5 = 15$ (Class V):

$$S_5 = \frac{5}{2}(3 + 15) = \frac{5}{2} \times 18 = 5 \times 9 = \boxed{\mathbf{45}\text{ trees}}$$

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★